INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY

AI Ethics, Governance and Legal Compliance Assessment

AICS-201 | AI Ethics, Governance & Legal Frameworks

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Github link: https://github.com/HaileamlakWudu/SecureFinance-AI-Ethics.git

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1. Introduction and System Overview

1.1 Background

Artificial intelligence can improve financial services through identity verification, fraud detection, transaction monitoring, and automated credit assessment. However, these applications introduce risks involving privacy, discrimination, financial exclusion, and incorrect automated decisions. This assessment examines these risks and recommends safeguards for responsible AI deployment at SecureFinance Africa.

1.2 SecureFinance Africa Case Study

SecureFinance Africa is a fictional financial technology company planning to use AI to improve financial services. Its initial deployment is proposed for Nigeria, with potential expansion into other African countries and international markets.

1.3 AI Systems under Assessment

The assessment focuses on three proposed AI systems:

- Identity Verification: Uses identity documents and biometric matching to verify customers.
- Fraud Detection: Analyses financial transactions to identify suspicious activities.
- Financial Risk Assessment: Evaluates customer creditworthiness and supports lending decisions.

1.4 Assessment Objectives and Scope

The assessment aims to identify ethical risks, evaluate AI governance practices, examine relevant legal requirements, and test fairness, explainability, and reliability using synthetic datasets.

It considers customers, financial institutions, employees, regulators, and technology providers. All practical experiments use synthetic data; therefore, the findings do not establish actual production performance or legal compliance.

2. Ethical Risk Assessment

2.1 Ethical Risk Identification

The following eight ethical risks were identified for SecureFinance Africa's proposed AI systems:

1. **Biometric Privacy Violations and Surveillance**: Misuse, unauthorised access, or excessive collection of sensitive biometric information, including facial images.

2. **Algorithmic Bias and Discrimination**: AI decisions may unfairly disadvantage certain groups based on gender, location, income, or socioeconomic background.

3. **Lack of Transparency and Explainability**: Customers may not understand why the AI rejects their identity, flags a transaction, or denies a loan.

4. **False Positives and False Negatives**: Incorrect AI predictions may block legitimate transactions or fail to detect fraudulent activities.

5. **Digital Exclusion and Unequal Access**: Customers with limited internet access, smartphones, digital literacy, or formal identification may struggle to use the services.

6. **Data Security and Unauthorised Access**: Personal and financial information may be exposed through cyberattacks, data breaches, or improper access.

7. **Insufficient Human Oversight and Accountability**: Excessive reliance on automated decisions may prevent customers from receiving meaningful human review or challenging incorrect decisions.

8. **Operational Reliability and Infrastructure Disruption**: Power outages, network failures, or system errors may interrupt AI services and affect customers' access to financial services.

2.2 Identification and Analysis of Ethical Risks

E01: Biometric Privacy Violations and Unethical Surveillance

1. What is the risk?

Biometric privacy violations involve the unauthorised or excessive collection, processing, storage or sharing of unique physical characteristics, including fingerprints, facial geometry and voiceprints. Unlike passwords, biometric characteristics cannot easily be changed if compromised. Their collection and processing therefore require strong privacy protections and responsible data governance.

A related concern is unethical surveillance and function creep. This occurs when biometric or video-analytics systems originally introduced for a limited purpose are later used to monitor, track or profile individuals beyond that purpose without appropriate justification or safeguards.

2. How could it occur?

SecureFinance Africa could introduce an AI-powered facial recognition feature in its mobile application to support customer identity verification and loan onboarding. The system might transmit facial templates to a third-party cloud provider to improve verification performance

without adequately informing customers about the processing or obtaining an appropriate legal basis.

If the third-party provider experienced a security breach, stored biometric information could be exposed.

A separate example involves the introduction of AI-powered video analytics in physical branches for queue management and identity verification. Management could later expand the system to monitor employee movements, analyse customer behaviour near advertisements or share foot-traffic information with commercial partners. Such secondary uses could exceed the original purpose for which the system was introduced.

3. Who is affected?

- Customers, particularly individuals who depend on digital financial services and may have limited understanding of complex privacy notices.
- Employees who could be subjected to unnecessary behavioural monitoring.
- Vulnerable populations who may be inaccurately profiled or treated unfairly based on automated behavioural analysis.
- SecureFinance Africa, which could experience reputational damage and regulatory consequences.

4. What is the potential harm?

- Exposure of biometric information could increase the risk of identity misuse and impersonation.
- Customers may lose confidence in the organization's ability to protect their personal information.
- Unnecessary monitoring could undermine privacy, personal autonomy and employee morale.
- Unauthorised secondary use of information could create legal, ethical and reputational consequences.

5. What should SecureFinance Africa do?

- Apply data minimisation by collecting only biometric information necessary for a clearly defined purpose.
- Assess whether biometric matching can be performed locally on a user's device rather than transmitting raw biometric information to central servers.
- Use appropriate legal grounds and provide clear, accessible explanations of biometric processing. Where consent is relied upon, it should be informed and meaningful.
- Protect stored biometric templates through encryption, secure key management, access restrictions and appropriate biometric-template protection methods.
- Establish strict purpose limitations that prevent the use of verification systems for unrelated behavioural profiling or advertising.
- Provide clear notices explaining video analytics, what information is collected and why.

- Conduct privacy and ethical impact assessments and periodically review third-party providers.
- Establish documented retention and deletion procedures consistent with the purpose of processing and applicable legal obligations.

E02: Algorithmic Bias and Discrimination

1. What is the risk?

Algorithmic bias occurs when an AI system produces systematically unfair outcomes for particular individuals or demographic groups. It can arise from unrepresentative training data, historical inequalities, inappropriate model design, biased input variables or inadequate testing.

In financial services, biased automated decisions may disadvantage individuals based on characteristics such as gender, geographical location, age or socioeconomic circumstances. This raises ethical concerns about fairness, equality and equal access to financial opportunities.

2. How could it occur?

SecureFinance Africa could deploy an automated credit-scoring model trained using historical financial records. If the training data reflects unequal access to formal banking, property ownership or employment opportunities, the model may learn patterns that disadvantage groups that were historically underrepresented.

For example, a financially stable female entrepreneur operating an informal business could receive a lower credit score than a similarly qualified applicant because the model relies heavily on formal employment history or registered assets. The resulting decision could be a loan rejection or less favourable borrowing conditions.

This is a hypothetical scenario illustrating how historical inequalities or unsuitable proxy variables could influence automated credit decisions. Actual bias would need to be established through testing of the model and its data.

3. Who is affected?

- Women and other groups that may be underrepresented in historical financial datasets.
- Rural populations and informal-sector workers.
- Young applicants and individuals with limited formal credit histories.
- Small businesses and entrepreneurs seeking access to finance.
- SecureFinance Africa, which could face reputational and regulatory consequences.

4. What is the potential harm?

- Qualified applicants may be unfairly denied access to credit.
- Existing economic inequalities may be reinforced through repeated automated decisions.
- Individuals may find it harder to build financial histories or access business opportunities.

- Discriminatory outcomes could damage public confidence and expose the organization to regulatory scrutiny.

5. What should SecureFinance Africa do?

- Assess whether training data adequately represents the populations and markets the system is intended to serve.
- Examine input variables for inappropriate proxies that could produce discriminatory outcomes.
- Conduct fairness testing across relevant demographic and socioeconomic groups, where lawful and appropriate data is available.
- Evaluate differences in approval rates, false rejection rates and other relevant performance measures across groups.
- Use synthetic data for controlled testing where appropriate, while recognising that synthetic data alone cannot establish fairness in real-world populations.
- Provide meaningful human review and an accessible appeal process for applicants who believe an automated decision is incorrect or unfair.
- Monitor model performance and fairness after deployment, including following changes to data, markets or model versions.
- Document identified biases, corrective actions and decisions to approve or restrict model deployment.

E03: Lack of Transparency and Explainability

1. What is the risk?

Lack of transparency and explainability occurs when an AI system makes important decisions without providing understandable information about the factors influencing its outputs.

In financial services, customers may be denied loans, assigned lower credit limits or flagged for suspicious activity without receiving a meaningful explanation. This creates accountability problems and makes it difficult for customers, employees and auditors to understand or challenge decisions.

2. How could it occur?

SecureFinance Africa could use a complex machine-learning model to determine customer credit limits. The model might process financial histories and other permitted alternative data to generate a credit assessment.

A long-term customer applying for an agricultural loan could receive a substantial reduction in their borrowing limit. However, frontline employees may be unable to explain the main factors behind the result because the model's operation and decision logic have not been adequately documented or made understandable.

The customer may consequently be unable to identify inaccurate information or understand what could be improved before making another application.

3. Who is affected?

- Borrowers and small businesses affected by automated lending decisions.
- Customer-service employees and loan officers responsible for explaining decisions.
- Compliance officers and internal auditors responsible for reviewing AI systems.
- SecureFinance Africa's management and governance teams.

4. What is the potential harm?

- Customers may lose trust in the organization.
- Individuals may be unable to identify or correct inaccurate information used in automated decisions.
- Unexplained decisions may make it difficult to challenge unfair outcomes.
- Internal auditors may struggle to establish whether the system operates as intended.
- The organization may fail to meet applicable transparency, accountability or explanation requirements.

5. What should SecureFinance Africa do?

- Document the purpose, design, limitations, inputs and intended use of each AI model.
- Prefer inherently interpretable models for high-impact decisions when they provide adequate performance.
- Evaluate explanation methods such as SHAP and LIME where complex models are used. These tools can support analysis but do not automatically guarantee that explanations are complete, faithful or understandable.
- Provide customers with clear explanations of significant automated decisions and the principal factors involved.
- Establish processes that allow customers to correct inaccurate information.
- Provide accessible channels for appeals and meaningful human review.
- Maintain decision logs and model documentation to support internal audits and accountability.

E04: False Positives and False Negatives

1. What is the risk?

False positives occur when a fraud-detection system incorrectly identifies legitimate activity as suspicious. False negatives occur when the system fails to identify actual fraudulent activity.

Both errors create ethical and operational problems. False positives can unfairly restrict innocent customers, while false negatives can expose customers and the organization to financial losses.

2. How could it occur?

SecureFinance Africa could introduce an AI-powered fraud-detection engine to monitor transactions in real time.

Two possible scenarios illustrate the risk:

- **False positive:** A rural merchant receives an unusually large payment from a regular customer. The system identifies the transaction as suspicious and automatically restricts the merchant's wallet. The merchant cannot access funds needed to purchase inventory while the transaction is reviewed.
- **False negative:** A fraud group carries out multiple small unauthorised transactions designed to resemble ordinary customer activity. The detection system fails to identify the pattern, allowing fraudulent transactions to continue.

3. Who is affected?

- Legitimate customers and merchants whose transactions are incorrectly restricted.
- Customers whose accounts are targeted by fraud that the system fails to detect.
- Fraud-monitoring and compliance employees responsible for investigating alerts.
- SecureFinance Africa, which may experience financial and reputational losses.

4. What is the potential harm?

- Legitimate customers may lose access to money needed for business activities and daily expenses.
- Undetected fraud may cause direct financial losses to customers and the organization.
- Excessive false alerts may overwhelm investigation teams and contribute to alert fatigue.
- Delayed responses to genuine fraud may increase the scale of financial harm.

5. What should SecureFinance Africa do?

- Establish proportionate, risk-based responses rather than automatically applying the same restriction to every suspicious transaction.
- Use appropriate verification steps for lower-risk alerts and escalate higher-risk cases for investigation.
- Regularly evaluate false-positive and false-negative rates using representative test data and operational feedback.
- Calibrate detection thresholds according to the organization's risk tolerance and legal obligations.
- Provide accessible, timely customer support and appeal channels.
- Ensure authorised employees can review and, where appropriate, override incorrect automated restrictions.
- Monitor alert volumes and investigation outcomes to identify alert fatigue and weaknesses in detection performance.

E05: Digital Exclusion and Unequal Access

1. What is the risk?

Digital exclusion occurs when people are unable to access or benefit from financial services because they lack suitable devices, reliable internet connectivity, digital literacy or the information required by an AI system.

This is an important ethical concern for financial services operating in markets where access to smartphones, affordable mobile data and reliable connectivity may differ significantly between communities.

2. How could it occur?

SecureFinance Africa could make its smartphone application the primary channel for customer onboarding and loan applications. Its AI-powered processes might depend on high-quality facial scans, continuous internet connectivity and smartphone-based behavioural information.

A rural farmer using a basic feature phone or living in an area with unreliable network coverage may be unable to complete the application. If no alternative process exists, the individual could be excluded from financial services or required to travel to a physical branch at additional cost.

3. Who is affected?

- Rural and low-income populations with limited access to smartphones or mobile data.
- Elderly customers and individuals with limited digital literacy.
- People living in areas with unreliable telecommunications infrastructure.
- Individuals with disabilities who may encounter inaccessible digital interfaces.
- SecureFinance Africa, which may fail to reach parts of its intended customer market.

4. What is the potential harm?

- Individuals may be excluded from formal banking and credit services.
- Existing inequalities between digitally connected and underserved communities may increase.
- Customers may have to spend additional time and money accessing alternative services.
- People excluded from formal financial services may become more dependent on informal or unregulated lending arrangements.
- The organization may fail to achieve its financial-inclusion objectives.

5. What should SecureFinance Africa do?

- Provide accessible alternatives to smartphone-only onboarding.
- Consider low-bandwidth applications and appropriate USSD or SMS channels for services that can safely be supported through them.
- Establish authorised agent-assisted or branch-assisted onboarding for customers who cannot complete digital processes independently.

- Avoid collecting unnecessary smartphone, location or behavioural information as a condition of accessing financial services.
- Design interfaces that support different languages, accessibility requirements and levels of digital literacy.
- Ensure that alternative onboarding routes apply appropriate identity and security checks.
- Monitor service access and completion rates across relevant customer groups to identify possible exclusion.

E06: Data Security and Unauthorised Access

1. What is the risk?

Data security and unauthorised access refer to the risk that sensitive information used or generated by AI systems may be exposed, altered, stolen or accessed by individuals without appropriate permission.

AI systems may process large quantities of personal, financial, biometric and transactional information. Weak access controls, insecure cloud configurations, vulnerable software or inadequate third-party security arrangements could expose this information.

2. How could it occur?

SecureFinance Africa could consolidate customer transaction histories, national identification information and mobile-wallet records in a centralised data environment to support model development and analysis.

An engineer might accidentally configure a cloud storage service to allow public access while troubleshooting a technical problem. If the misconfiguration remains undetected, an unauthorised person could discover and download sensitive records.

This is an illustrative security scenario, not a claim that an actual breach has occurred.

3. Who is affected?

- Customers whose personal and financial information is exposed.
- Employees and authorised users whose access credentials may be compromised.
- SecureFinance Africa's management, technology and security teams.
- Third-party service providers responsible for storing or processing information.

4. What is the potential harm?

- Exposed information could be used for targeted phishing, identity misuse and financial fraud.
- Customers may experience financial losses and long-term privacy consequences.
- The organization may face regulatory investigations, financial penalties and remediation costs.

- Loss of customer confidence may damage the organization's reputation.
- A security incident could disrupt financial services and business operations.

5. What should SecureFinance Africa do?

- Apply least-privilege access controls and role-based access management.
- Require multi-factor authentication for sensitive systems and administrative accounts.
- Encrypt sensitive information in transit and at rest, with appropriate key management.
- Use network segmentation, secure cloud configurations and continuous security monitoring.
- Use synthetic or appropriately de-identified data in development and testing environments where possible.
- Log and review access to sensitive production information.
- Assess third-party providers before granting access to customer information.
- Conduct vulnerability assessments and controlled penetration testing.
- Maintain incident-response, backup and recovery procedures.
- Establish processes for investigating and reporting personal-data breaches in accordance with applicable requirements.

E07: Insufficient Human Oversight and Accountability

1. What is the risk?

Insufficient human oversight occurs when an organization relies excessively on AI-generated recommendations or decisions without meaningful human evaluation, accountability or opportunities for affected individuals to seek review.

Automation bias can contribute to this problem. It occurs when employees place excessive confidence in automated outputs and accept recommendations without adequately questioning their accuracy.

2. How could it occur?

SecureFinance Africa could use an AI-powered transaction-monitoring system to identify potentially suspicious transactions. The system may generate a large number of alerts for the compliance team.

As alert volumes increase, employees may begin approving or escalating recommendations without properly examining the available evidence. In a hypothetical example, the system could incorrectly flag a humanitarian organization's transfer because of a misunderstood transaction description. Employees might repeatedly reject the organization's appeals because they assume the AI-generated result must be correct.

3. Who is affected?

- Customers and businesses whose accounts or transactions are incorrectly restricted.

- Humanitarian organizations and other institutions that depend on timely financial transfers.
- Compliance employees responsible for reviewing automated alerts.
- SecureFinance Africa's management and governance teams.

4. What is the potential harm?

- Innocent customers may lose access to funds or experience unnecessary service restrictions.
- Humanitarian, business or personal activities may be disrupted.
- Employees may become overly dependent on AI recommendations and lose critical decision-making capacity.
- Unclear responsibility may make it difficult to investigate and correct harmful decisions.
- Customer confidence may decline when meaningful review and appeals are unavailable.

5. What should SecureFinance Africa do?

- Define clear responsibilities for the development, approval, deployment and monitoring of AI systems.
- Establish risk-based human oversight for significant decisions and actions.
- Require appropriate human review for high-impact decisions, such as major lending decisions or prolonged account restrictions, subject to applicable legal and operational requirements.
- Ensure that reviewers have sufficient information, training and authority to challenge or override AI recommendations.
- Provide accessible customer appeal and escalation channels.
- Record important decisions, human interventions and overrides for audit purposes.
- Monitor employee workloads and alert volumes to reduce automation bias and alert fatigue.
- Ensure that accountability remains with the organization and its responsible personnel rather than being attributed solely to the AI system or its vendor.

E08: Operational Reliability and Infrastructure

1. What is the risk?

Operational reliability and infrastructure risks arise when AI systems fail because of software defects, network outages, power interruptions, insufficient computing capacity, poor maintenance or inadequate disaster-recovery arrangements.

For financial services, system availability is particularly important because customers and merchants may depend on digital platforms for payments, access to funds and credit-related services.

2. How could it occur?

SecureFinance Africa could host its transaction-monitoring and credit-assessment systems on centralised cloud infrastructure. A regional power interruption or telecommunications outage might disrupt connectivity to the primary service.

If backup systems and failover procedures were inadequate, customers could be unable to complete transactions or access some financial services. A software update or model deployment could also introduce a defect that interrupts transaction processing.

The consequences could be especially serious during periods of high transaction activity. This scenario is illustrative; the actual likelihood and scale of disruption depend on the infrastructure and resilience controls selected by the organization.

3. Who is affected?

- Small-scale merchants and businesses that depend on digital payments.
- Customers who need access to financial services for everyday transactions.
- Employees responsible for technical support and incident response.
- SecureFinance Africa's operational, technology and management teams.

4. What is the potential harm?

- Customers may temporarily lose access to financial services.
- Merchants may lose sales or experience delays in receiving payments.
- Transactions may remain incomplete or require reconciliation.
- System interruptions may damage customer confidence.
- Poorly managed recovery could introduce further operational or security problems.

5. What should SecureFinance Africa do?

- Design resilient infrastructure with appropriate redundancy and backup arrangements.
- Establish disaster-recovery and business-continuity plans.
- Test network, power and cloud-service failover procedures.
- Use controlled software releases, model validation and rollback procedures.
- Establish monitoring and alerting to identify service degradation quickly.
- Define safe degraded operating modes for periods when central AI services are unavailable.
- Ensure that queued transactions can be reconciled securely after service restoration.
- Conduct controlled resilience and recovery testing in suitable test environments.
- Establish clear responsibilities and communication procedures for operational incidents.

2.3 Risk Prioritisation Method

The eight identified ethical risks were prioritised according to two factors: likelihood and impact.

The assessment uses a three-point scale:

Score	Likelihood or impact
1	Low
2	Medium
3	High

The overall priority score is calculated using:

Risk Priority = Likelihood × Impact

The following priority categories are used:

Priority score	Classification
1–2	Low
3–4	Medium
6	High
9	Critical

The scores represent preliminary qualitative judgements based on the proposed AI use cases and the illustrative scenarios. They are not statistical probabilities or measured model-performance results. They should be reviewed when more information about SecureFinance Africa's actual systems, controls, operational environment and testing results becomes available.

2.4 Risk Prioritisation Matrix

ID	Ethical risk	Likelihood	Impact	Priority score	Classification
E01	Biometric privacy and surveillance	2	3	6	High
E02	Algorithmic bias and discrimination	3	3	9	Critical
E03	Lack of transparency and explainability	3	2	6	High
E04	False positives and false negatives	3	2	6	High
E05	Digital exclusion and unequal access	3	3	9	Critical
E06	Data security and unauthorised access	2	3	6	High
E07	Insufficient human oversight and accountability	2	2	4	Medium

E08	Operational reliability and infrastructure	3	3	9	Critical

2.4.1 Justification of Risk Scores

Critical-priority risks

E02 – Algorithmic Bias and Discrimination (3 × 3 = 9)

The likelihood is assessed as high because credit-scoring systems trained on historical financial information may reproduce existing inequalities if the training data, input variables and model outputs are not adequately assessed. The impact is also high because unfair credit decisions could affect access to loans and business opportunities across multiple customer groups. This risk requires attention before the credit-assessment system is deployed.

E05 – Digital Exclusion and Unequal Access (3 × 3 = 9)

The likelihood is assessed as high if SecureFinance Africa makes smartphone-based applications, continuous connectivity or extensive digital information necessary for accessing its services. The impact is high because customers without suitable devices, reliable internet access or digital skills could be excluded from financial services. This could undermine the organization's intended financial-inclusion objectives. The likelihood should be reassessed once the actual onboarding channels and customer access requirements are defined.

E08 – Operational Reliability and Infrastructure (3 × 3 = 9)

The likelihood is assessed as high because financial platforms depend on reliable power, telecommunications, software and hosting infrastructure, all of which may experience interruptions or technical failures. The impact is high because outages could prevent customers and merchants from completing important transactions or accessing services. Redundancy, recovery arrangements and safe degraded operating procedures are therefore important pre-deployment requirements.

High-priority risks

E01 – Biometric Privacy and Surveillance (2 × 3 = 6)

The likelihood is assessed as medium because the extent of exposure depends on the biometric architecture, storage arrangements, third-party integrations and privacy controls selected by the organization. The impact is high because compromised biometric information may create long-term privacy and identity-misuse risks, while function creep could undermine individual autonomy and trust.

E03 – Lack of Transparency and Explainability (3 × 2 = 6)

The likelihood is assessed as high where complex models are used without adequate documentation, explanation mechanisms or customer communication. The impact is medium because unexplained decisions can undermine trust, prevent customers from correcting inaccurate information and make internal accountability more difficult. The severity could be higher where an unexplained decision has a substantial effect on an individual's access to financial services.

E04 – False Positives and False Negatives (3 × 2 = 6)

The likelihood is assessed as high because fraud-detection systems may incorrectly classify legitimate and fraudulent transactions, particularly when customer behaviour changes or detection thresholds are not appropriately calibrated. The impact is medium because proportionate interventions and effective investigation procedures may reduce harm. However, serious cases involving prolonged restrictions or substantial undetected fraud could have higher impacts.

E06 – Data Security and Unauthorised Access (2 × 3 = 6)

The likelihood is assessed as medium because the actual level of exposure depends on the organization's data architecture, access controls, security monitoring and third-party arrangements, none of which are fully established in the case information. The impact is high because unauthorised access to combined identity and financial information could create significant privacy, fraud and regulatory consequences.

Medium-priority risk

E07 – Insufficient Human Oversight and Accountability (2 × 2 = 4)

The likelihood is assessed as medium because automation bias and excessive reliance on AI recommendations may arise when employees handle large volumes of automated alerts or lack appropriate training and review procedures. The impact is assessed as medium because effective escalation, human review and accountability arrangements could limit the consequences. However, the impact may be high in cases involving prolonged account restrictions, serious financial harm or other high-impact decisions.

2.5 Overall Ethical Assessment

2.5.1 Three Most Serious Ethical Risks

Based on the prioritisation matrix, the three most serious ethical risks for SecureFinance Africa are:

1. E02 – Algorithmic Bias and Discrimination.
2. E05 – Digital Exclusion and Unequal Access.

3. E08 – Operational Reliability and Infrastructure.

Each received a priority score of 9, representing high preliminary likelihood and high potential impact.

These risks are particularly important because they affect the fairness, accessibility and reliability of the organization's proposed financial services.

E02: Algorithmic Bias and Discrimination

A biased credit-assessment system could unfairly disadvantage women, rural populations, informal-sector workers and other groups. If the model reproduces historical inequalities, customers who could otherwise qualify for financial services may receive unfairly low assessments or loan rejections. This could undermine equal access to financial opportunities and damage public confidence in the organization.

E05: Digital Exclusion and Unequal Access

A financial platform that depends excessively on smartphones, reliable internet connectivity or advanced digital skills may exclude customers who cannot meet those requirements. This is especially important for a financial service intended to reach underserved communities. Without accessible alternatives, the organization could reinforce the same barriers to financial inclusion that it intends to reduce.

E08: Operational Reliability and Infrastructure

Financial services depend on reliable systems. If the organization's AI platform becomes unavailable because of power interruptions, connectivity failures or software defects, customers and merchants may be unable to access important services. Inadequate recovery arrangements could increase the duration and consequences of disruptions, affecting livelihoods and customer trust.

2.5.2 Why These Risks Must Be Addressed Before Deployment

These risks must be considered before deployment because their consequences could affect large numbers of customers and undermine the organization's ability to deliver fair, accessible and dependable financial services.

First, deploying a credit model without appropriate fairness testing could produce discriminatory outcomes. Once the system is used in real lending decisions, affected customers may lose access to credit and business opportunities. Correcting the problem after deployment may require reviewing previous decisions, modifying the model and addressing customer complaints. Pre-deployment testing is therefore important.

Second, launching a platform without alternative access channels could exclude customers who lack smartphones, affordable mobile data or reliable connectivity. This would conflict with the

objective of extending financial services to underserved populations. Accessibility must therefore be considered during system design rather than treated as an optional improvement after launch.

Third, launching a financial platform without adequate resilience arrangements could expose customers and merchants to service interruptions. A disruption could prevent transactions from being completed and create difficulties in accessing funds. Testing backup systems, recovery procedures and safe operating modes before deployment can reduce these risks.

Together, these three risks could undermine the organization's financial-inclusion objectives, customer relationships and operational sustainability. Addressing them before deployment is therefore an important part of responsible AI governance.

2.5.3 Minimum Ethical Safeguards Before Go-Live

SecureFinance Africa should establish the following minimum safeguards before deploying its AI systems.

1. Pre-deployment fairness testing and bias mitigation

The organization should assess the representativeness and suitability of its training data and evaluate model outcomes across relevant demographic and socioeconomic groups. Testing should examine approval rates, false rejection rates and other appropriate fairness measures. Synthetic data may support controlled testing, but it should not be treated as proof that a model is fair in real-world use. Identified problems should be addressed and retested before the system is approved for deployment.

2. Accessible onboarding and alternative service channels

SecureFinance Africa should provide appropriate alternatives for customers who cannot use smartphone-based services. These may include low-bandwidth applications, suitable USSD or SMS services, and authorized agent-assisted or branch-assisted onboarding. The organization should ensure that alternative channels maintain appropriate security and identity-verification requirements.

3. Operational resilience and disaster recovery

The organization should implement appropriate infrastructure redundancy, backup arrangements, disaster-recovery procedures and business-continuity plans. It should test failover and recovery arrangements under controlled conditions. Safe degraded operating procedures should define which services can continue during an outage and how transactions will be securely reconciled after normal operations resume.

4. Meaningful human oversight and customer appeals

High-impact decisions should have appropriate human oversight. Employees responsible for reviewing AI recommendations must receive adequate training and have the authority to challenge or override incorrect outputs. Customers should have accessible channels to request explanations, correct inaccurate information and appeal decisions. Review times and escalation procedures should be defined according to the urgency and nature of the decision.

5. Data protection and information security

The organization should apply data minimisation, least-privilege access, appropriate encryption, secure key management and multi-factor authentication. Sensitive information should be protected throughout collection, storage, processing and transmission. Development and testing environments should use synthetic or appropriately de-identified information where possible. Biometric information should be processed and retained only where necessary, with clear purpose limitations and appropriate safeguards.

6. Transparency and explainability

SecureFinance Africa should document the purpose, limitations and operation of its AI models. Customers affected by significant automated decisions should receive understandable explanations of the principal factors influencing those decisions. Tools such as SHAP and LIME may assist with technical explanations, but their outputs should be evaluated before being used to communicate decisions to customers. Explanation and appeal processes should be accessible and understandable.

2.5.4 Conclusion

The ethical risk assessment identifies eight important risks associated with SecureFinance Africa's proposed use of artificial intelligence in financial services. These include biometric privacy and surveillance, algorithmic bias, lack of transparency, fraud-detection errors, digital exclusion, data security weaknesses, insufficient human oversight and operational reliability.

The prioritization exercise identifies algorithmic bias and discrimination, digital exclusion, and operational reliability as the three highest-priority risks. Each received a preliminary score of 9 based on high assessed likelihood and high potential impact.

The assessment demonstrates that responsible AI deployment requires more than technical accuracy. SecureFinance Africa must also consider fairness, privacy, transparency, accessibility, accountability and system resilience. Implementing the recommended safeguards before deployment, followed by continuous monitoring and periodic reassessment, will help the organization reduce potential harm and build a more trustworthy AI-enabled financial platform.

3. NIST AI Risk Management Framework

3.1 Overview of NIST AI RMF 1.0

The NIST AI Risk Management Framework (AI RMF 1.0) provides a voluntary framework for organizations to identify, assess, manage and monitor risks associated with AI systems.

It has four core functions:

1, GOVERN – Establish accountability

Set policies, responsibilities, organizational culture and oversight for responsible AI.

2, MAP – Understand the context

Identify the AI system's purpose, stakeholders, operating environment and potential risks.

3, MEASURE – Assess the risks

Analyse, test and evaluate risks, including fairness, reliability, security, privacy and transparency.

4, MANAGE – Treat and monitor risks

Prioritize risks, select responses, assign responsibilities and monitor risks throughout the AI lifecycle.

These functions work together; they are not simply four steps that an organization completes once. Risk management should continue throughout the AI system's lifecycle. [5]

3.2 GOVERN: Establishing Organizational AI Governance

The GOVERN function of the NIST AI Risk Management Framework (AI RMF 1.0) establishes the organizational structures, policies, responsibilities, and accountability mechanisms needed to manage AI-related risks throughout the AI lifecycle. For SecureFinance Africa, this function is particularly important because its proposed AI applications—identity verification, fraud detection, and financial risk assessment—may influence access to financial services, the handling of sensitive personal information, and decisions affecting customers.

The governance arrangements below are recommendations for SecureFinance Africa. They should not be interpreted as evidence that the organization already has these committees, departments, or controls.

3.2.1 AI Governance Policy

SecureFinance Africa should establish a formal Responsible AI Framework Policy covering the development, procurement, deployment, monitoring, and retirement of AI systems. The policy should establish minimum requirements for fairness, transparency, privacy, security, reliability, human oversight, and accountability.

An Algorithmic Impact Assessment (AIA) should be required for proposed AI projects before development or procurement proceeds. The assessment should identify the intended purpose, affected stakeholders, types of data involved, foreseeable harms, potential discrimination, and necessary safeguards. It should be revisited before deployment and following significant changes to the model, data, purpose, or operating environment.

This process would help ensure that ethical and operational risks are considered early rather than addressed only after a system has caused harm.

3.2.2 Roles and Responsibilities

A cross-functional AI Governance Committee should be established to coordinate responsible AI practices and review the risks associated with proposed and deployed AI systems. The committee should include representatives from risk management, data protection, compliance, information security, business operations, and AI development.

The proposed responsibilities are:

- **Board Risk Committee:** Provide strategic oversight, approve the organization's AI risk appetite, and review significant AI risks and incidents.
- **AI Governance Committee:** Review AI impact assessments, assess whether proposed controls are adequate, and make or recommend approval decisions according to delegated authority.
- **Chief Risk Officer (CRO):** Coordinate the integration of AI risks into the organization's wider risk-management framework.
- **Data Protection Officer (DPO):** Advise on personal-data protection obligations and oversee relevant privacy risks.
- **AI/Model Development Team:** Develop, document, validate, and monitor models, including testing for fairness, performance, and reliability.
- **Information Security Team:** Protect AI systems, infrastructure, credentials, and sensitive information against unauthorised access and other security threats.
- **Compliance Team:** Monitor applicable legal and regulatory requirements and coordinate customer complaints and escalations.
- **Business/Model Owners:** Remain responsible for the approved business purpose, operational use, and day-to-day monitoring of the AI systems under their control.

These responsibilities should be documented in an approved governance charter, with clear reporting and escalation arrangements.

3.2.3 Accountability and Customer Recourse

SecureFinance Africa should establish an accountability structure that does not treat an algorithm or an external vendor as a substitute for organizational responsibility.

Business and Model Owners should be accountable for the operational use of their systems, including ensuring that the systems are used for approved purposes and that adverse outcomes

are escalated. Executive management should provide organizational oversight and ensure that appropriate resources and controls are in place. The Board should oversee significant risks in accordance with its governance responsibilities.

A designated AI Appeals and Grievance function, potentially located within the Compliance function, should receive and investigate customer complaints involving AI-assisted decisions. Customers should have an accessible way to challenge decisions that may affect account access, identity verification, fraud flags, or financial assessments. The process should provide for appropriate human review, correction of inaccurate information, and escalation where necessary.

The organization's legal responsibilities and the responsibilities of individual officers must be determined under applicable law; they should not be assigned automatically to the CRO or CEO without a legal basis.

3.2.4 Risk Culture and Employee Training

SecureFinance Africa should develop an organizational culture in which employees understand that AI outputs require appropriate scrutiny and that responsibility remains with people and the organization.

Mandatory AI risk-awareness training should be provided to relevant personnel, with quarterly workshops proposed as a suitable training schedule. Training should be tailored to job responsibilities:

- **Frontline and customer-service personnel:** Recognise automation bias, understand when an AI result requires escalation, and explain available complaint or review channels.
- **AI developers and data analysts:** Identify potential demographic bias, data-quality problems, model limitations, and inappropriate uses of data.
- **Branch and operations personnel:** Recognise unauthorised data collection, excessive data use, and potential function creep.
- **Managers and decision-makers:** Understand AI risk reporting, approval requirements, and their accountability for systems used in their business areas.

Training completion and policy compliance should be monitored. The organization may also incorporate relevant governance responsibilities into employee performance assessments.

3.2.5 Third-Party and Vendor Management

Because AI services may depend on external providers, including cloud platforms, biometric verification services, and third-party APIs, SecureFinance Africa should establish a formal vendor risk-assessment process.

Before engaging a provider, the organization should assess its security practices, privacy protections, data-handling arrangements, subcontractors, incident-response procedures, and

ability to meet contractual requirements. Independent security audit reports or equivalent assurance evidence should be requested where appropriate.

Contracts and Service Level Agreements should clearly address:

- Permitted purposes for processing and restrictions on secondary use of data.
- Security and privacy obligations.
- Breach notification and incident-cooperation requirements.
- Subcontracting and data-location arrangements.
- Audit and assurance rights.
- Data retention, return, and deletion at the end of the service.
- Responsibility, remedies, and liability for contractual breaches.

Continuous monitoring of vendor access and periodic reassessment should complement initial procurement checks. Contract termination should be available for serious or unresolved violations, subject to contractual obligations, legal requirements, and a safe transition plan.

3.2.6 Ongoing Oversight and Reporting

AI governance should continue after a system has been approved and deployed. The proposed AI Governance Committee should conduct regular reviews of model performance, data and model drift, false-positive and false-negative trends, customer disputes, security incidents, and changes in the system's operating environment.

Monthly reviews are proposed for systems with significant customer or operational impact. Review frequency should otherwise be proportionate to the system's risk and the rate at which it changes.

The committee should provide a biannual AI Health and Compliance Report to the Board Risk Committee. The report should summarise significant risks, incidents, customer complaints, model performance, outstanding corrective actions, vendor issues, and changes requiring management attention.

The organization should also establish escalation procedures for serious incidents, including the authority to restrict, suspend, or withdraw an AI system where its risks cannot be adequately controlled.

3.2.7 Approval and Enforcement Arrangements

The proposed approval process should use risk-based delegation:

- The AI Governance Committee should review impact assessments and recommend approval conditions.
- Delegated executive management should approve lower-risk systems within the limits of the organization's approved AI risk policy.

- The Board Risk Committee should approve or oversee high-impact and high-risk AI deployments and material changes, in accordance with the organization's governance charter.

For all risk levels, deployment should be blocked until the required assessments, testing, documentation, security checks, and approvals have been completed.

Employee compliance should be supported through mandatory training, policy acknowledgement, management review, and automated deployment controls that prevent unapproved models or code from entering production. Third-party compliance should be supported through contractual requirements, access monitoring, periodic assurance reviews, and corrective-action or termination procedures for serious violations.

3.2.8 GOVERN Function Assessment

The proposed GOVERN framework provides SecureFinance Africa with a foundation for managing AI risks through defined policies, responsibilities, approval processes, employee training, vendor oversight, customer recourse, and continuing management review.

Its effectiveness would depend on implementation. The organization would need to formally approve the governance policy, appoint responsible personnel, establish the committee and reporting arrangements, document delegated approval authority, and maintains evidence that the proposed controls operate in practice.

At this assessment stage, these are recommended governance arrangements rather than verified controls already implemented by SecureFinance Africa..

3.3 MAP: Understanding the AI System Context

The MAP function of the NIST AI Risk Management Framework (AI RMF 1.0) is concerned with understanding the context in which an AI system is developed and used. This includes its intended purpose, affected stakeholders, operating environment, data requirements, potential impacts, and limitations. Mapping these factors helps an organisation identify risks before deployment and determine which risks require further measurement and management.

For SecureFinance Africa, the following analysis considers three proposed AI applications: identity verification, fraud detection, and financial risk assessment. The architecture and operating procedures described below are proposed design assumptions and have not been independently verified as implemented systems.

3.3.1 Intended Purpose and System Boundaries

A. Identity Verification System

The proposed identity verification system is intended to authenticate customers by comparing government identification information with real-time facial matching results. Its purpose is to support customer onboarding and identity checks.

The proposed failure procedure is to flag a profile for human review, with a target verification period of 24 hours. During review, high-value withdrawals may be restricted while standard account access remains available. This is a proposed operational policy, not a confirmed service-level commitment. The organisation would need to define what constitutes a high-value withdrawal, how restrictions are authorised, and how customers can challenge an incorrect rejection.

The system's boundaries should include the identity data supplied by customers, any government registry integrations, biometric processing services, matching decisions, and the human review process.

B. Fraud Detection System

The proposed fraud detection system is intended to identify potentially unauthorised transactions, account takeover attempts, and suspicious transaction patterns, including unusually frequent low-value transactions.

It is expected to analyse transaction activity in real time. Under the proposed failure procedure, a suspicious transaction would be temporarily suspended, a one-hour wallet hold could be applied, and the customer would receive a verification notification. These actions are proposed controls and require further definition and testing before deployment. In particular, the organisation should determine when a transaction may be held, when a wallet restriction is proportionate, how customers are authenticated, and how legitimate transactions can be restored promptly.

The system boundary should include transaction data, account and device information, detection outputs, temporary restrictions, customer notifications, and human investigation or escalation.

C. Financial Risk Assessment System

The proposed financial risk assessment system is intended to evaluate customer creditworthiness and recommend or determine borrowing limits for micro-loans.

The proposed design distinguishes between lower-value loans, which may be processed automatically within an approved threshold, and higher-value loans, for which the AI system provides recommendations for human or authorised decision-maker review. The exact threshold and approval authority have not yet been established.

The system boundary should include the data used to assess creditworthiness, model-generated risk scores or recommendations, lending-limit decisions, human review where applicable, and subsequent repayment monitoring.

3.3.2 Stakeholder Identification

The proposed AI systems may affect several groups:

- **Customers and borrowers:** Individuals applying for digital accounts, identity verification, or micro-loans. They may be affected by inaccurate identity matches, fraud flags, or credit assessments.
- **Merchants:** Businesses using digital wallets or payment services. Incorrect fraud alerts or account restrictions could interrupt their ability to receive or use payments.
- **Rural and underserved populations:** Customers with limited access to formal identification, reliable internet, smartphones, or mobile data may face additional barriers to digital financial services.
- **Frontline employees:** Staff responsible for onboarding, customer support, and handling disputes may need to interpret AI outputs and escalate questionable decisions.
- **Risk analysts:** Personnel responsible for investigating suspicious transactions and reviewing model-generated alerts.
- **Management and governance bodies:** Business owners, risk management, compliance, and executive leadership responsible for approving and overseeing the use of AI.
- **External service providers:** Biometric verification vendors, cloud providers, telecommunications partners, and payment infrastructure providers may support system operations.
- **Regulators:** Relevant financial and data-protection authorities may have oversight responsibilities depending on the services offered and the applicable legal framework.

The stakeholder analysis indicates that AI-related impacts extend beyond the direct user of a system. For example, a false fraud alert may affect a merchant's ability to conduct business, while an inaccessible identity verification process may prevent an otherwise eligible customer from accessing financial services.

3.3.3 Proposed Data Architecture

The proposed data requirements differ across the three AI applications.

Identity verification data

- National identity card information.
- Live facial images captured during verification.
- Information obtained from government identity registries, where authorised and available.

Possible additional data includes utility bills and residential geolocation information. These are not confirmed requirements and should only be collected if necessary, proportionate, and supported by an appropriate legal basis.

Fraud detection data

- Transaction amounts and timestamps.
- Sending and receiving wallet identifiers.

- Device-related information, such as device hardware signatures.

Possible additional data includes social-network relationships and application interaction patterns, such as scrolling or typing behaviour. These are optional examples rather than confirmed data sources. Their use would require additional assessment of necessity, privacy, security, fairness, and the risk of collecting more information than required.

Financial risk assessment data

- Historical micro-loan repayment performance.
- Internal deposit volumes.
- Wallet balance patterns over time.

Possible additional indicators include mobile airtime purchase histories and local agricultural yield forecasts. Their relevance and reliability would need to be tested before they are used in lending decisions. In particular, alternative data should not be assumed to be an accurate or fair representation of an individual's ability or willingness to repay.

The organisation should maintain documentation identifying the source, purpose, quality, retention period, access permissions, and permitted uses of each data category.

3.3.4 Operating Environment

The proposed digital ecosystem is expected to use lightweight Android applications, USSD interfaces, central payment gateways, and a multi-cloud hosting arrangement. These channels are intended to support customers with different levels of device capability and internet access.

The operating environment may include areas affected by unreliable electricity and intermittent network connectivity. Such conditions could affect real-time identity checks, transaction monitoring, customer notifications, and access to lending services.

The proposed deployment roadmap begins with a Nigeria pilot, with possible future expansion into West and East African markets. Expansion would require separate assessments of local regulatory requirements, available identity infrastructure, data quality, language and accessibility needs, and market-specific financial conditions. The initial Nigerian design should not automatically be assumed suitable for every future country.

The specific cloud providers, payment integrations, biometric vendors, and government registry arrangements remain unconfirmed.

3.3.5 Potential Impacts of AI Failures

Identity verification

A false rejection may prevent a legitimate customer from completing onboarding or accessing particular account functions. This could disproportionately affect people whose identity

documents are difficult to verify or whose facial images are affected by image quality, lighting, or other technical limitations. A false acceptance may allow an impostor to pass an identity check and potentially expose customers or the organisation to fraud and compliance risks.

Fraud detection

A false positive may cause a legitimate transaction to be suspended or a customer's wallet to be temporarily restricted. For merchants, such restrictions could interrupt normal business activity. A false negative may allow unauthorised transactions or account takeover activity to proceed without detection.

Financial risk assessment

An inaccurate or biased credit assessment may result in unfair rejection, inappropriate borrowing limits, or unequal access to micro-loans. If the model underestimates repayment risk, customers may receive borrowing limits they cannot reasonably manage, potentially contributing to over-indebtedness and financial losses for the organisation. The potential for gender-related exclusion, including impacts on female entrepreneurs, should be tested rather than assumed to have already occurred.

These impacts demonstrate why model performance must be assessed alongside the consequences of decisions for customers and other affected stakeholders.

3.3.6 Assumptions, Uncertainties, and Limitations

Several important design decisions remain unresolved:

1. **Model architecture:** The specific algorithms, model structures, and development methods have not yet been selected.
2. **Third-party providers:** The biometric, cloud, and other external service providers have not been identified.
3. **Government registry availability:** The availability, reliability, access conditions, and response times of national identity databases have not been verified.
4. **Data completeness and quality:** The completeness, accuracy, representativeness, and permitted use of proposed data sources remain to be assessed.
5. **Automation thresholds:** The exact boundary between automated decisions and human review has not been formally established.
6. **Operational restrictions:** The proposed 24-hour identity review period and one-hour fraud-related wallet hold have not been validated or approved.
7. **Geographic expansion:** The requirements for future deployment outside Nigeria remain subject to country-specific assessment.

These uncertainties limit the conclusions that can currently be drawn about the systems' actual performance, security, fairness, and operational readiness. They should be resolved or documented as explicit assumptions before deployment.

3.3.7 MAP Function Assessment

The MAP analysis identifies the proposed purposes, stakeholders, data categories, operating environment, potential harms, and important uncertainties associated with SecureFinance Africa's three AI applications.

The analysis highlights several issues requiring further evaluation: biometric data protection, the proportionality of fraud-related restrictions, the fairness of alternative credit indicators, the accessibility of digital services, and the reliability of external infrastructure.

The next stage should test these concerns using documented methods and available evidence. At present, the system descriptions are proposed designs, not evidence of actual deployed capabilities or measured outcomes.

3.4 MEASURE: Testing and Evaluation of AI Risks

The MEASURE function of the NIST AI Risk Management Framework (AI RMF 1.0) focuses on assessing, analysing, and monitoring AI risks through appropriate testing and evaluation. For SecureFinance Africa, the evaluation considers three proposed AI applications: identity verification, fraud detection, and financial risk assessment.

A Python-based evaluation was conducted using synthetic datasets. The reported dataset sizes were 1,000 identity verification records, 2,000 fraud transactions, and 1,000 financial applicants. A random seed of 42 was used to support reproducibility, and the stated training/test split for the trained models was 75%/25%.

Identity verification was evaluated using a simulated matching score and threshold rather than a trained biometric recognition model. Fraud detection and credit risk assessment used StandardScaler and Logistic Regression. Permutation importance based on ROC-AUC was used for the reported credit model explainability analysis. Controlled input perturbations were applied to examine prediction stability.

All results are synthetic evaluation evidence. They do not represent production performance or prove that SecureFinance Africa has deployed these systems.

3.4.1 Identity Verification Evaluation

The identity verification evaluation used a simulated matching score with a threshold of 65 out of 100. A legitimate identity was defined as the positive class.

The reported overall accuracy was 87.00%, with a false acceptance rate of 0% and a false rejection rate of 15.13%.

The gender comparison produced the following results:

Metric	Female	Male
Records	525	475
Accuracy	87.62%	86.32%
False acceptance rate	0%	0%
False rejection rate	14.25%	16.13%

The difference in false rejection rates indicates that legitimate female identities were rejected less frequently than legitimate male identities in this synthetic evaluation. The difference was approximately 1.87 percentage points.

Regional results also varied. Kano recorded the highest false rejection rate at 19.44%, followed by Abuja at 16.15%, Lagos at 15.21%, Rivers at 13.27%, and Other at 11.80%.

These results indicate that verification outcomes can differ across demographic and geographic groups. The zero false acceptance rate is encouraging within the generated dataset, but it should not be interpreted as proof that the system can prevent all identity fraud.

Identity reliability testing

The reliability test introduced controlled changes by reducing image quality by five points and network quality by three points.

The reported results were:

- Mean absolute prediction change: 0.079139.
- Prediction correlation: 0.999518.
- Decision change rate: 14.8%.

Although prediction correlation remained high, the decision change rate indicates that some verification outcomes changed following the simulated input modifications. This is important because a small change in input conditions could affect whether a legitimate customer passes verification.

The results suggest that further testing of image quality, connectivity conditions, and decision thresholds is necessary. The synthetic test does not establish how the system would perform with real biometric images or actual network conditions.

3.4.2 Fraud Detection Evaluation

The fraud detection model used StandardScaler and Logistic Regression. Fraudulent transactions were defined as the positive class.

The overall results were:

Metric	Result
Accuracy	89.20%
Precision	42.11%
Recall	15.69%
F1-score	22.86%
False-positive rate	2.45%
False-negative rate	84.31%
Average simulated detection time	112.38 ms

The most important finding is the low recall of 15.69%, together with a false-negative rate of 84.31%. This indicates that the model missed a substantial proportion of fraudulent transactions in the synthetic evaluation.

Although the accuracy was 89.20%, accuracy alone provides an incomplete assessment of fraud detection performance. Where legitimate transactions substantially outnumber fraudulent transactions, a model may achieve relatively high accuracy while failing to identify many actual fraudulent transactions.

The precision of 42.11% also indicates that fewer than half of the transactions flagged as fraudulent were actually fraudulent in the evaluation data. This creates a risk of unnecessary customer verification and transaction restrictions.

Fraud fairness by gender

The reported gender results were:

Metric	Female	Male
Transactions	273	227
Accuracy	91.94%	85.90%
Precision	33.33%	57.14%
Recall	22.22%	12.12%
False-positive rate	3.14%	1.55%
False-negative rate	77.78%	87.88%

The model had a higher recall for female-associated transactions, while male-associated transactions had a higher false-negative rate. Female-associated transactions also had a higher false-positive rate. These differences should be investigated to determine whether they arise from the synthetic data, the model, or the selected detection threshold.

Fraud fairness by region

The regional results showed substantial variation:

Region	Recall	False-negative rate
Abuja	0%	100%
Kano	8.33%	91.67%
Lagos	12.50%	87.50%
Other	8.33%	91.67%
Rivers	41.67%	58.33%

Abuja recorded no detected fraudulent transactions in the reported regional evaluation, while Rivers had the highest recall at 41.67%. This suggests that the model's detection performance was not consistent across the tested regional groups.

These regional results should be interpreted cautiously because the number of fraudulent examples within each group may be small. Group-level performance should be examined alongside the number of actual fraud cases and the corresponding confidence intervals.

Fraud reliability testing

The fraud model was tested by increasing transaction amounts by 2% and reducing the time since the previous transaction by two minutes.

The reported results were:

- Mean absolute prediction change: 0.000543.
- Prediction correlation: 0.999978.
- Decision change rate: 0.4%.

These results indicate high prediction stability under the specific perturbations applied. However, prediction stability does not compensate for the low fraud recall. A model can consistently produce predictions while consistently missing fraudulent activity.

The reported average detection time of 112.38 milliseconds was simulated and should not be treated as a measured production response time.

3.4.3 Credit Risk Assessment Evaluation

The credit risk assessment model used StandardScaler and Logistic Regression. Default was defined as the positive class, with a default prediction threshold of 0.30.

The reported gender results were:

Metric	Female	Male
Applicants	128	122
Approval rate	67.97%	72.13%
Accuracy	85.16%	81.15%

False-positive rate	13.04%	16.49%
False-negative rate	19.44%	28.00%
Actual default rate	28.13%	20.49%

The female approval rate was approximately 4.16 percentage points lower than the male approval rate. The actual default rate was also higher among female applicants in the synthetic dataset.

Since default was the positive class, a false positive represents a non-defaulter incorrectly classified as a defaulter, while a false negative represents an actual defaulter incorrectly classified as a non-defaulter. These definitions are important when interpreting the potential consequences of model errors.

The results show differences in approval rates and classification errors across gender groups. However, the results alone do not establish discriminatory treatment. Further investigation would need to examine the generated data, relevant financial characteristics, and the model's decision process.

Credit fairness by region

Region	Applicants	Approval rate	False-positive rate
Abuja	62	75.81%	8.51%
Kano	57	70.18%	15.91%
Lagos	40	60.00%	19.23%
Other	33	66.67%	19.23%
Rivers	58	72.41%	15.22%

Lagos recorded the lowest approval rate at 60.00%, while Abuja recorded the highest at 75.81%. The difference between the highest and lowest regional approval rates was approximately 15.81 percentage points.

The regional false-positive rate difference was approximately 10.72 percentage points, while the false-negative rate difference was approximately 12.38 percentage points.

These differences provide evidence of variation in the model's reported outcomes across regions. They justify further fairness testing, particularly to determine whether regional patterns in the synthetic data or the model's use of financial indicators are influencing lending decisions.

3.4.4 Credit Model Explainability

Permutation importance using ROC-AUC was applied to examine the relative influence of input features on the credit model's predictive performance.

The reported feature contribution percentages were:

Feature	Contribution
Loan amount	73.66%
Income	16.88%
Debt ratio	5.44%
Credit history years	3.77%
Employment years	0.25%
Age	0.002%

Loan amount was the most influential feature, accounting for approximately 73.66% of the reported normalised contribution. Income was the second most influential feature at 16.88%. Age had the smallest reported contribution.

This result suggests that the model's predictive performance was particularly sensitive to loan amount in the synthetic evaluation. SecureFinance Africa should investigate whether loan amount is receiving disproportionate influence compared with indicators such as income, debt ratio, and repayment history.

Permutation importance measures the effect of disrupting individual features on model performance. It does not establish that a feature causes a particular decision or that the model is fair. Correlated features and the data-generation process may also affect the results.

3.4.5 Credit Model Reliability and Stability

The credit model was tested by reducing income by 2%, increasing loan amount by 2%, and increasing the debt ratio by 0.01.

The reported results were:

- Mean absolute prediction change: 0.013533.
- Prediction correlation: 0.999528.
- Decision change rate: 1.2%.

These results suggest that predictions were highly correlated under the tested changes, with 1.2% of decisions changing. This provides preliminary evidence of stability under the specific perturbation scenario.

However, stability under a limited set of controlled changes does not demonstrate that the model is reliable across different economic conditions, customer populations, or changes in the quality of input data.

3.4.6 Comparative Assessment of the Three AI Systems

The evaluation identifies different risk concerns across the three proposed systems:

System	Main finding	Risk requiring attention
Identity verification	87.00% accuracy and 15.13% false rejection rate	Incorrect rejection and sensitivity to input changes
Fraud detection	15.69% recall and 84.31% false-negative rate	Failure to detect fraudulent transactions
Credit risk assessment	15.81 percentage-point regional approval-rate range	Unequal outcomes and differences in classification errors

The fraud detection system presents the most immediate performance concern because its reported recall is low. The identity verification system requires additional testing of false rejection and stability under image and network changes. The credit risk assessment system requires further investigation of differences in approval rates and error rates across demographic and regional groups.

These findings support the need for additional validation and risk controls before any real-world deployment.

3.4.7 Limitations of the Evaluation

Several limitations affect the interpretation of the results:

1. All datasets and labels were synthetically generated. They do not represent actual SecureFinance Africa customers or transactions.
2. Identity verification used a simulated score-based method rather than a trained biometric recognition model.
3. Fraud detection and credit risk assessment used Logistic Regression with StandardScaler; the findings may differ under other models or real-world data.
4. Fraud detection times were simulated rather than measured in a production environment.
5. Fairness differences are diagnostic indicators and do not automatically establish discrimination.
6. Permutation importance identifies changes in predictive performance, not causal relationships.
7. Controlled perturbation tests do not establish reliability under all real-world operating conditions.
8. The results depend on the synthetic data-generation process, class distribution, selected thresholds, and evaluation methods.
9. The evaluation does not establish production accuracy, operational readiness, or legal compliance.

The evaluation summary reports a 75%/25% training/test split. The fraud and credit subgroup tables contain 500 and 250 records respectively, consistent with the reported test-set proportions.

The identity subgroup table contains all 1,000 records, however, so its evaluation population should be confirmed and documented separately because identity verification used a simulated score-based method.

Further validation should include confusion-matrix counts, the number of actual positive cases in each subgroup, and confidence intervals for important fairness metrics. These additions would help establish how much confidence can be placed in the reported differences.

3.4.8 MEASURE Function Assessment

The MEASURE function has been addressed through synthetic testing of identity verification, fraud detection, and credit risk assessment. The results provide preliminary evidence of differences in model outcomes, classification errors, feature importance, and prediction stability.

The most significant finding is the low fraud detection recall, which indicates that the proposed fraud model would require substantial improvement before it could be considered suitable for operational use. Identity verification and credit assessment also require additional testing to investigate rejection rates, decision stability, and differences across demographic and regional groups.

The results are useful for identifying potential risks and demonstrating an evaluation methodology. They should not be interpreted as proof of real-world fairness, accuracy, security, or reliability.

3.5 MANAGE – Risk Response and Continuous Improvement

The MANAGE function of the NIST AI Risk Management Framework focuses on prioritising identified risks, selecting appropriate responses, assigning responsibility, and continuously monitoring AI systems. For SecureFinance Africa, the following actions are proposed based on the synthetic evaluation results.

1. Improve fraud detection performance

The fraud model achieved 15.69% recall and an 84.31% false negative rate, meaning it failed to identify many fraudulent transactions in the synthetic test data. SecureFinance should review the training data, investigate missed fraud cases, and evaluate alternative models and decision thresholds. The model should not be approved for production use until its fraud detection performance has been substantially improved and independently tested.

2. Improve identity verification reliability

The identity verification simulation showed a 14.8% decision change rate under controlled input perturbations. SecureFinance should investigate the sensitivity of its verification threshold and improve the consistency of image and network-quality handling. Uncertain or failed verification

results should be referred for human review rather than automatically causing permanent customer rejection.

3. Investigate credit assessment disparities

The credit model showed a 4.16-percentage-point gender approval gap and a 15.81-percentage-point difference across regions. SecureFinance should investigate the causes of these differences, review the influence of loan amount and income, and test the model using representative data. Credit decisions should remain subject to appropriate human review and a customer appeal process.

4. Establish continuous monitoring and accountability

A proposed AI Governance Committee should oversee these improvements, with model owners responsible for implementation and the Compliance and Risk functions providing independent oversight. Monitoring should include fraud recall, false positive rates, identity decision changes, credit approval differences, model drift, customer complaints, and security incidents. Significant deterioration should trigger investigation, corrective action, or temporary suspension of the affected model.

5. Apply controlled deployment and reassessment

SecureFinance should begin with controlled testing before allowing high-impact automated decisions in production. Models should be reassessed after significant updates, changes in data, or changes in operating conditions. Existing controls and residual risks must be documented before deployment approval.

Conclusion: The synthetic evaluation identifies issues that require further investigation, particularly weak fraud detection, identity verification instability, and credit approval disparities. SecureFinance should address these risks before production deployment and maintain continuous monitoring throughout each AI system's lifecycle. The current synthetic results are not sufficient to establish production readiness.

4. Legal and Regulatory Compliance Assessment

4.2 Nigeria Data Protection Act (NDPA) 2023

The Nigeria Data Protection Act 2023 is the primary data protection legislation relevant to SecureFinance's proposed Nigerian operations. It establishes requirements for lawful, fair, transparent, and secure processing of personal data.

SecureFinance will process sensitive information, including identity documents, facial images, financial records, and transaction histories. Under Section 30, sensitive personal data is subject to additional processing requirements. The company should establish an appropriate lawful basis, provide clear privacy information, minimise collected data, and define retention periods.

Section 28 requires a Data Privacy Impact Assessment (DPIA) where processing is likely to create high risks to individuals. Given SecureFinance's proposed biometric processing, profiling, and automated financial decisions, a DPIA should be completed before deployment.

Section 37 addresses automated decision-making that produces legal or similarly significant effects. SecureFinance should provide appropriate safeguards, including human intervention, opportunities for customers to express their views, and the ability to contest decisions.

The company must also implement appropriate security measures under Section 39 and comply with Sections 41–43 when transferring personal data outside Nigeria. [1]

4.3 European Union Artificial Intelligence Act

The EU AI Act, Regulation (EU) 2024/1689, establishes risk-based requirements for AI systems. Although SecureFinance initially plans to operate in Nigeria, the Act may become relevant if its systems are placed on the EU market, supplied or used within the EU, or their outputs are used in the EU under the Act's territorial provisions.

The credit-risk assessment system requires particular attention because AI systems evaluating the creditworthiness of individuals are generally classified as high-risk under Annex III. The Act provides an exception for systems used to detect financial fraud.

Biometric systems also require careful classification. Certain remote biometric identification and biometric categorisation systems can be high-risk, while biometric verification used solely to confirm a person's claimed identity is excluded from that specific Annex III category.

Where a system is classified as high-risk, relevant requirements include risk management, data governance, technical documentation, record-keeping, transparency, human oversight, accuracy, robustness, and cybersecurity.

SecureFinance should therefore conduct a formal EU AI Act applicability and classification assessment before entering European markets. [2]

4.4 UNESCO Recommendation on the Ethics of AI

UNESCO's 2021 Recommendation provides an international ethical framework for responsible AI development and use. It promotes human rights, fairness, privacy, transparency, accountability, and human oversight.

These principles are relevant to SecureFinance because inaccurate identity verification, biased credit assessments, and incorrect fraud alerts could negatively affect customers. The company should provide understandable explanations, accessible review procedures, and safeguards against discrimination and exclusion.

The Recommendation is an international ethical standard, not a directly binding Nigerian statute. [3]

4.5 OECD AI Principles

The OECD AI Principles, initially adopted in 2019 and updated in May 2024, promote trustworthy and human-centred AI. Their five values-based principles cover:

1. Inclusive growth, sustainable development, and well-being.
2. Human rights, fairness, privacy, and democratic values.
3. Transparency and explainability.
4. Robustness, security, and safety.
5. Accountability.

These principles support SecureFinance's proposed fairness testing, customer explanations, security controls, human review, and continuous monitoring. They are non-binding international recommendations rather than direct legal obligations on the company. [4]

4.6 Overall Compliance Assessment and Recommendations

SecureFinance should prioritise the following actions before deployment:

- Complete a DPIA covering biometric data, financial profiling, and automated decisions.
- Document lawful processing bases and provide clear privacy notices.
- Establish data retention, deletion, security, and access-control policies.
- Provide human review and customer appeal mechanisms for significant automated decisions.
- Assess cross-border data transfers and third-party data processors.
- Evaluate fairness, accuracy, explainability, and cybersecurity throughout the AI lifecycle.
- Conduct a separate legal assessment before entering additional countries or the EU market.

Conclusion: The Nigeria Data Protection Act is directly relevant to SecureFinance's proposed Nigerian operations. The EU AI Act may apply if the company's activities meet its territorial and material scope requirements. UNESCO and OECD provide additional ethical guidance. SecureFinance should address these requirements before deployment and maintain ongoing compliance reviews as its systems and geographical operations develop.

5. Practical Implementation and Evidence

5.1 Experimental Environment

The experiments were implemented using Python. The analysis used Pandas and NumPy for data generation and processing, Scikit-learn for machine learning and evaluation, and Matplotlib for visualisation. The experiments were performed locally using a Python development environment.

5.2 Synthetic Dataset Description

Three synthetic datasets were generated for the evaluation.

Dataset	Records	Purpose
Identity verification	1,000	Identity evaluation
Fraud transactions	2,000	Fraud detection
Credit applicants	1,000	Credit assessment

No real customer or financial data was used.

5.3 Implementation Methodology

The securefinance_measure.py script generates the synthetic datasets and performs the evaluation. It simulates identity verification using matching scores and trains Logistic Regression models for fraud detection and credit assessment.

The evaluation includes accuracy, precision, recall, false positive and false negative rates, and subgroup fairness comparisons across gender and region. Permutation importance is used to examine credit model feature importance, while controlled input perturbations are used to assess prediction stability.

5.4 Experimental Results

The experiments produced the following key results:

Evaluation metric	Result
Identity verification accuracy	87.00%
Identity decision change rate	14.80%
Fraud detection accuracy	89.20%
Fraud detection recall	15.69%
Fraud false negative rate	84.31%
Gender credit approval difference	4.16 percentage points
Regional credit approval difference	15.81 percentage points

The generated dashboard (summery_dashboard.png) provides a visual summary of the evaluation results. [AG]

5.5 Results Interpretation

The experiments highlight three important concerns: weak fraud detection recall, identity verification instability under controlled input changes, and differences in credit approval rates across demographic groups.

These findings support the need for further fairness testing, model improvement, and reliability validation. However, because the experiments use synthetic data, they do not prove actual discrimination or establish real-world model performance.

5.6 Reproducibility

The experiments can be reproduced by installing the required Python libraries and executing:

```
python analysis/securefinance_measure.py
```

The script generates the synthetic datasets and evaluation outputs stored in the datasets/ and results/ directories. The complete implementation and supporting evidence are available in the project's GitHub repository.

6. Integrated Recommendations and Conclusion

6.1 Priority Recommendations

SecureFinance should prioritise the following actions before deployment:

1. Improve fraud detection recall and investigate missed fraudulent transactions.
2. Improve identity verification stability and introduce human review for uncertain results.
3. Investigate gender and regional differences in credit approval.
4. Complete data protection impact assessments and implement appropriate security controls.
5. Establish clear AI governance, accountability, customer appeals, and continuous monitoring.

6.2 Proposed Implementation Roadmap

Phase	Proposed activity
Phase 1	Establish governance policies, complete DPIA, and assess risks.
Phase 2	Improve and retest AI models.
Phase 3	Validate security, fairness, and reliability.
Phase 4	Conduct controlled pilot deployment after required approvals.
Phase 5	Continuously monitor performance, risks, and compliance.

6.3 Conclusion

This assessment identified significant ethical, technical, operational, and legal risks associated with SecureFinance Africa's proposed AI systems. The synthetic evaluation highlighted weaknesses in fraud detection, identity verification stability, and differences in credit approval outcomes.

SecureFinance should not proceed directly to full production deployment based on these experimental results. It should first improve model performance, validate fairness, strengthen data protection, establish meaningful human oversight, and complete the necessary legal and security assessments.

Ultimately, responsible AI deployment requires continuous monitoring, transparency, accountability, and protection of customer rights throughout the AI lifecycle.

7. References

1. Nigeria Data Protection Commission. (2023). Nigeria Data Protection Act 2023. https://ndpc.gov.ng/wp-content/uploads/2024/03/Nigeria_Data_Protection_Act_2023.pdf
2. European Union. (2024). Regulation (EU) 2024/1689 – Artificial Intelligence Act. https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1721145089347&uri=CELEX:32024R1689
3. UNESCO. (2021). Recommendation on the Ethics of Artificial Intelligence. https://www.unesco.org/en/legal-affairs/recommendation-ethics-artificial-intelligence
4. OECD. (2019, updated 2024). OECD AI Principles. https://oecd.ai/en/ai-principles
5. NIST AI Risk Management Framework 1.0 https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf

8. Appendices

Appendix A – Complete Eight-Risk Register

Located on risk_register/risk_register.csv

Appendix B – Synthetic Dataset Descriptions

Dataset	Records	Description
identity_verification.csv	1,000	Synthetic identity verification records
fraud_transactions.csv	2,000	Synthetic financial transaction records
credit_applicants.csv	1,000	Synthetic credit applicant

records

All records are artificially generated and contain no real customer information.

Appendix C – Python Implementation Methodology

Include:

- securefinance_measure.py.

Appendix D – Complete Identity Evaluation Results

Include the actual outputs:

- identity_metrics.csv
- identity_fairness.csv
- identity_regional_fairness.csv
- identity_reliability.csv

These provide the detailed identity verification accuracy, FAR, FRR, subgroup results, and reliability measurements.

Appendix E – Complete Fraud Evaluation Results

Include:

- fraud_metrics.csv
- fraud_fairness.csv
- fraud_regional_fairness.csv
- fraud_reliability.csv

These provide the fraud detection accuracy, precision, recall, F1-score, false positive rate, false negative rate, subgroup results, and reliability measurements.

Appendix F – Complete Credit Evaluation Results

Include:

- credit_metrics.csv
- credit_gender_fairness.csv
- credit_region_fairness.csv
- credit_approval_differences.csv
- credit_explainability.csv
- credit_reliability.csv

These provide credit model performance, approval differences, subgroup fairness, feature importance, and reliability results.

Appendix G – Dashboard and Supporting Screenshots

